

# COMP0083: Kernel Methods - Lecture Notes

## Comprehensive Course Notes

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## Part I

# Foundations: Hilbert Spaces and Inner Products

## 1 Motivation and Overview

Kernels are fundamental tools in machine learning that allow us to:

- Compare objects based on their features without explicitly computing feature representations
- Control model smoothness and generalization to avoid overfitting
- Extend linear methods to non-linear settings through feature space transformations

The key insight is that many algorithms depend only on inner products between data points, which we can compute implicitly through a kernel function.

## 2 Inner Products and Norms

### 2.1 Inner Product Spaces

**Definition 2.1** (Inner Product). Let  $H$  be a vector space over  $\mathbb{R}$ . A function  $\langle \cdot, \cdot \rangle_H : H \times H \rightarrow \mathbb{R}$  is an inner product if it satisfies:

1. **Linearity in the first argument:**

$$\langle \alpha f + \beta g, h \rangle_H = \alpha \langle f, h \rangle_H + \beta \langle g, h \rangle_H$$

for all  $\alpha, \beta \in \mathbb{R}$  and  $f, g, h \in H$ .

2. **Symmetry:**  $\langle f, g \rangle_H = \langle g, f \rangle_H$  for all  $f, g \in H$ .

3. **Positive Definiteness:**  $\langle f, f \rangle_H \geq 0$  for all  $f \in H$ , with equality if and only if  $f = 0$ .

The inner product induces a norm on  $H$ :

$$\|f\|_H := \sqrt{\langle f, f \rangle_H}$$

### 2.2 Norms on Vector Spaces

**Definition 2.2** (Norm). Let  $F$  be a vector space over  $\mathbb{R}$ . A function  $\|\cdot\|_F : F \rightarrow [0, \infty)$  is a norm if it satisfies:

1. **Separation of points:**  $\|f\|_F = 0 \Leftrightarrow f = 0$
2. **Positive homogeneity:**  $\|\lambda f\|_F = |\lambda| \|f\|_F$  for all  $\lambda \in \mathbb{R}$  and  $f \in F$
3. **Triangle inequality:**  $\|f + g\|_F \leq \|f\|_F + \|g\|_F$  for all  $f, g \in F$

Every normed vector space carries an induced metric:  $d(f, g) = \|f - g\|_F$ . Conversely, every inner product space is a normed vector space with the norm induced by the inner product.

## 2.3 Inner Products on $\mathbb{R}^n$

**Definition 2.3** (Inner Product on  $\mathbb{R}^n$ ). A symmetric positive definite matrix  $M \in \mathbb{R}^{n \times n}$  defines an inner product on  $\mathbb{R}^n$ :

$$\langle x, y \rangle_M := x^T M y \quad \text{for all } x, y \in \mathbb{R}^n$$

The special case  $M = I$  gives the standard Euclidean inner product:  $\langle x, y \rangle = x^T y$ .

## 3 Hilbert Spaces and Completeness

### 3.1 Cauchy Sequences and Completeness

A sequence  $\{f_n\}_{n=1}^\infty$  is Cauchy with respect to the induced metric if for every  $\varepsilon > 0$ , there exists  $N$  such that for all  $n, m > N$ :

$$\|f_n - f_m\|_F < \varepsilon$$

Every convergent sequence is Cauchy, but not all Cauchy sequences converge. For example, the sequence  $1, 1.4, 1.41, 1.414, \dots$  is Cauchy in  $\mathbb{Q}$  but converges (to  $\sqrt{2}$ ) only in  $\mathbb{R}$ .

**Definition 3.1** (Complete Metric Space). A normed vector space  $F$  is complete if every Cauchy sequence  $\{f_n\}$  in  $F$  converges to some  $f \in F$  (i.e.,  $\lim_{n \rightarrow \infty} \|f_n - f\|_F = 0$ ).

**Definition 3.2** (Hilbert Space). A Hilbert space is a complete inner product space. That is, it is a vector space with an inner product where every Cauchy sequence converges.

More formally, a Banach space is a complete normed vector space, while a Hilbert space is a complete inner product space.

### 3.2 Orthogonality and Orthonormal Basis

**Definition 3.3** (Orthogonal Complement). For a set  $M \subseteq H$  in an inner product space, its orthogonal complement is:

$$M^\perp := \{g \in H : \langle f, g \rangle = 0 \text{ for all } f \in M\}$$

The orthogonal complement is always a closed linear subspace. If  $M$  is a linear subspace, then  $M \cap M^\perp = \{0\}$  (discrete intersection).

**Definition 3.4** (Orthonormal Basis). An orthonormal basis of a Hilbert space  $H$  is a family  $\{e_k\}_{k \in B}$  of elements satisfying:

1. **Orthogonality:**  $\langle e_i, e_j \rangle = 0$  for all  $i \neq j$
2. **Normalization:**  $\|e_k\|_H = 1$  for all  $k \in B$
3. **Completeness:** The linear span of  $\{e_k\}$  is dense in  $H$  (closedness condition)

Such a basis allows every element  $f \in H$  to be represented as:

$$f = \sum_{k \in B} \langle f, e_k \rangle e_k$$

with convergence in the Hilbert space norm.

### 3.3 Important Properties in Inner Product Spaces

**Theorem 3.1** (Cauchy-Schwarz Inequality). For any elements  $f, g$  in an inner product space, we have:

$$|\langle f, g \rangle| \leq \|f\| \|g\|$$

with equality if and only if  $f$  and  $g$  are linearly dependent.

**Theorem 3.2** (Pythagorean Theorem). If  $f \perp g$  (i.e.,  $\langle f, g \rangle = 0$ ), then:

$$\|f + g\|^2 = \|f\|^2 + \|g\|^2$$

**Theorem 3.3** (Parallelogram Law).

$$2\|f\|^2 + 2\|g\|^2 = \|f + g\|^2 + \|f - g\|^2$$

## 4 Linear Operators

### 4.1 Bounded and Continuous Operators

**Definition 4.1** (Linear Operator). A map  $A : F \rightarrow G$  between vector spaces is a linear operator if:

$$A(\alpha_1 f_1 + \alpha_2 f_2) = \alpha_1 A(f_1) + \alpha_2 A(f_2)$$

for all scalars  $\alpha_1, \alpha_2$  and elements  $f_1, f_2 \in F$ .

When  $G = \mathbb{R}$  (or  $\mathbb{C}$ ), a linear operator is called a functional.

**Definition 4.2** (Operator Norm). The operator norm of a bounded linear operator  $A : F \rightarrow G$  is:

$$\|A\| := \sup_{f \in F, f \neq 0} \frac{\|Af\|_G}{\|f\|_F} = \sup_{\|f\|_F \leq 1} \|Af\|_G$$

If  $\|A\| < \infty$ , we say  $A$  is a bounded linear operator.

**Theorem 4.1** (Equivalence of Boundedness and Continuity). For a linear operator  $A : F \rightarrow G$  between normed spaces, the following are equivalent:

1.  $A$  is a bounded operator
2.  $A$  is continuous on all of  $F$
3.  $A$  is continuous at a single point in  $F$

### 4.2 The Riesz Representation Theorem

**Theorem 4.2** (Riesz Representation Theorem). Let  $H$  be a Hilbert space and let  $f : H \rightarrow \mathbb{R}$  be a bounded linear functional. Then there exists a unique  $y \in H$  such that:

$$f(x) = \langle x, y \rangle_H \quad \text{for all } x \in H$$

Moreover,  $\|f\|_{H'} = \|y\|_H$ , where  $\|f\|_{H'}$  is the operator norm of  $f$  on the dual space.

**Remark 4.1.** This theorem is fundamental: it says that every bounded linear functional on a Hilbert space can be represented as an inner product with some fixed vector. The mapping  $J : H \rightarrow H'$  defined by  $J(y)(x) = \langle x, y \rangle_H$  is an isometric isomorphism between  $H$  and its dual space  $H'$ .

## Part II

# Kernels and Reproducing Kernel Hilbert Spaces

## 5 Kernels: Definition and Properties

### 5.1 Kernel Definition and Positive Definiteness

**Definition 5.1** (Kernel). Let  $X$  be a non-empty set. A function  $k : X \times X \rightarrow \mathbb{R}$  is a kernel if there exists a Hilbert space  $H$  and a feature map  $\phi : X \rightarrow H$  such that:

$$k(x, x') = \langle \phi(x), \phi(x') \rangle_H \quad \text{for all } x, x' \in X$$

Equivalently, a kernel can be characterized through positive definiteness.

**Definition 5.2** (Positive Definite Kernel). A function  $k : X \times X \rightarrow \mathbb{R}$  is positive definite if for any finite set of points  $\{x_1, \dots, x_n\} \subseteq X$  and coefficients  $\{c_1, \dots, c_n\} \subseteq \mathbb{R}$ :

$$\sum_{i,j=1}^n c_i c_j k(x_i, x_j) \geq 0$$

The key insight is that kernels and positive definite functions are equivalent concepts.

### 5.2 Constructing Kernels

**Proposition 5.1** (Operations Preserving Kernels). The following operations preserve the kernel property:

1. **Positive scaling:** If  $k$  is a kernel and  $\alpha > 0$ , then  $\alpha k$  is a kernel.
2. **Sums of kernels:** If  $k_1$  and  $k_2$  are kernels on  $X$ , then  $k_1 + k_2$  is a kernel on  $X$ .
3. **Products of kernels:** If  $k_1$  is a kernel on  $X_1$  and  $k_2$  is a kernel on  $X_2$ , then the product

$$k(x, x') := k_1(x_1, x'_1) \cdot k_2(x_2, x'_2)$$

is a kernel on  $X_1 \times X_2$ .

4. **Kernel composition:** If  $A : X \rightarrow \tilde{X}$  is a map and  $\tilde{k}$  is a kernel on  $\tilde{X}$ , then  $k(x, x') := \tilde{k}(A(x), A(x'))$  is a kernel on  $X$ .

**Remark 5.1.** Not all differences of kernels are kernels. Additionally, a single kernel can admit infinitely many different feature map representations.

### 5.3 Examples of Kernels

**Definition 5.3** (Polynomial Kernel). The polynomial kernel of degree  $d$  is defined as:

$$k(x, x') := (\langle x, x' \rangle + c)^d$$

where  $d$  is a positive integer and  $c \geq 0$  is a constant. This kernel corresponds to the feature space of all monomials of degree up to  $d$  in the original features.

**Definition 5.4** (Gaussian (RBF) Kernel). The Gaussian or Radial Basis Function (RBF) kernel is:

$$k(x, x') := \exp(-\gamma \|x - x'\|^2)$$

where  $\gamma > 0$  is the bandwidth parameter. This kernel is universal and characteristic for probability measures.

**Definition 5.5** (Kernels from Power Series). For  $r \in (0, \infty]$  and coefficients  $\{a_n\}_{n=0}^{\infty}$  with  $a_n \geq 0$ , if the radius of convergence allows, we can define:

$$k(x, x') := \sum_{n=0}^{\infty} a_n \langle x, x' \rangle^n$$

for  $\langle x, x' \rangle \in (-r, r)$ .

## 6 Reproducing Kernel Hilbert Spaces (RKHS)

### 6.1 Evaluation Functionals

**Definition 6.1** (Evaluation Functional). Given a space  $H$  of functions  $f : X \rightarrow \mathbb{R}$  and a fixed point  $x_0 \in X$ , the evaluation functional at  $x_0$  is:

$$E_{x_0} : H \rightarrow \mathbb{R}, \quad E_{x_0}(f) := f(x_0)$$

This is a linear functional, but it is not always bounded with respect to every norm on  $H$ .

### 6.2 RKHS Definition

**Definition 6.2** (Reproducing Kernel Hilbert Space). A Hilbert space  $H$  of functions  $f : X \rightarrow \mathbb{R}$  is a Reproducing Kernel Hilbert Space (RKHS) if for every  $x \in X$ , the evaluation functional  $E_x : H \rightarrow \mathbb{R}$  defined by  $E_x(f) = f(x)$  is bounded.

Equivalently, there exists a family  $\{k_x\}_{x \in X} \subseteq H$  such that:

$$f(x) = \langle f, k_x \rangle_H \quad \text{for all } x \in X, f \in H$$

where  $k_x(y) = k(y, x)$  and  $k : X \times X \rightarrow \mathbb{R}$  is the reproducing kernel.

This definition encodes the **reproducing property**: function values can be recovered via inner products with kernel sections.

### 6.3 Moore-Aronszajn Theorem

**Theorem 6.1** (Moore-Aronszajn). There is a one-to-one correspondence between positive definite kernels on  $X \times X$  and Reproducing Kernel Hilbert Spaces on  $X$ .

Specifically, for every positive definite kernel  $k$  on  $X \times X$ , there exists a unique RKHS  $H_k$  with reproducing kernel  $k$ . Conversely, every RKHS has a unique reproducing kernel.

The RKHS  $H_k$  is the completion of the pre-Hilbert space spanned by  $\{k_x\}_{x \in X}$  with inner product:

$$\left\langle \sum_i \alpha_i k_{x_i}, \sum_j \beta_j k_{x_j} \right\rangle_{H_k} := \sum_{i,j} \alpha_i \beta_j k(x_i, x_j)$$

Key consequences:

- Every positive definite kernel defines a unique RKHS
- The feature map  $\phi(x) = k(\cdot, x)$  embeds  $X$  into  $H_k$
- The kernel value is  $k(x, x') = \langle \phi(x), \phi(x') \rangle_{H_k}$

### 6.4 Properties of RKHS and Smoothness

**Proposition 6.1** (Smoothness from Small RKHS Norm). If  $f \in H_k$  with small  $\|f\|_{H_k}$ , then  $f$  is smooth. Specifically:

$$|f(x) - f(x')| = |\langle f, k_x - k_{x'} \rangle| \leq \|f\|_{H_k} \|k_x - k_{x'}\|_{H_k}$$

For smooth kernels,  $\|k_x - k_{x'}\|_{H_k}$  is small when  $x$  is close to  $x'$ .

## 7 Spectral Properties and Function Representations

### 7.1 Integral Linear Operators and Mercer's Theorem

**Definition 7.1** (Integral Operator). Given a kernel  $k$  and a measure  $\nu$  on  $X$ , the integral operator is:

$$(T_k f)(x) := \int_X k(x, y) f(y) d\nu(y)$$

This is a linear operator mapping appropriate function spaces.

**Theorem 7.1** (Mercer's Theorem). Let  $k$  be a continuous kernel on a compact metric space  $X$ , and let  $\nu$  be a finite Borel measure with  $\text{supp}(\nu) = X$ . Then  $k$  admits an eigenfunction expansion:

$$k(x, y) = \sum_{j=1}^{\infty} \lambda_j e_j(x) e_j(y)$$

where  $\lambda_1 \geq \lambda_2 \geq \dots \geq 0$  are eigenvalues of the integral operator  $T_k$  and  $\{e_j\}$  are the corresponding orthonormal eigenfunctions. The series converges absolutely and uniformly on  $X \times X$ .



## 7.2 Spectral Theorem and Decomposition

**Theorem 7.2** (Spectral Decomposition). Let  $H$  be a Hilbert space and  $T : H \rightarrow H$  a compact, self-adjoint operator. Then there exists an at-most countable orthonormal set  $\{e_j\}_{j \in J}$  and eigenvalues  $\lambda_1 \geq \lambda_2 \geq \dots$  converging to zero, such that:

$$f = \sum_{j \in J} \langle f, e_j \rangle e_j \quad \text{and} \quad Tf = \sum_{j \in J} \lambda_j \langle f, e_j \rangle e_j$$

for all  $f \in H$ .

This allows us to represent any function in  $H_k$  and understand the decay of the spectrum.

## Part III

# Applications of Kernels

## 8 Kernel Methods in Machine Learning

### 8.1 Representer Theorem

**Theorem 8.1** (Representer Theorem). Consider a regularized empirical risk minimizer:

$$\min_{f \in H_k} \frac{1}{n} \sum_{i=1}^n L(y_i, f(x_i)) + \lambda \|f\|_{H_k}^2$$

where  $L$  is any loss function and  $\lambda > 0$ . The optimal solution has the form:

$$f^*(x) = \sum_{i=1}^n \alpha_i^* k(x, x_i)$$

That is, the optimal function is expressible as a linear combination of kernel evaluations at the training points.

This is why kernel methods are computationally practical: we only need to optimize over  $n$  coefficients rather than solving in infinite-dimensional spaces.

### 8.2 Kernel Ridge Regression

**Definition 8.1** (Kernel Ridge Regression). Kernel Ridge Regression solves:

$$\min_{f \in H_k} \frac{1}{n} \sum_{i=1}^n (y_i - f(x_i))^2 + \lambda \|f\|_{H_k}^2$$

By the representer theorem, the solution is:

$$f^*(x) = \sum_{i=1}^n \alpha_i^* k(x, x_i)$$

where  $\alpha^* = (K + n\lambda I)^{-1} \mathbf{y}$  and  $K$  is the kernel matrix with entries  $K_{ij} = k(x_i, x_j)$ .

## 9 Kernel Mean Embeddings

### 9.1 Definition and Properties

**Definition 9.1** (Kernel Mean Embedding). For a probability measure  $P$  on  $\mathcal{X}$  and kernel  $k$ , the kernel mean embedding is:

$$\mu_P := \mathbb{E}_{X \sim P}[k(\cdot, X)] = \int k(\cdot, x) dP(x) \in H_k$$

For an empirical measure  $P_n = \frac{1}{n} \sum_{i=1}^n \delta_{x_i}$ , the empirical mean embedding is:

$$\hat{\mu}_{P_n} := \frac{1}{n} \sum_{i=1}^n k(\cdot, x_i)$$

Mean embeddings map probability distributions to function space, preserving distributional information.

**Proposition 9.1** (Injectivity for Characteristic Kernels). A kernel  $k$  is called characteristic for probability measures on  $\mathcal{X}$  if the embedding  $P \mapsto \mu_P$  is injective. That is:

$$\mu_P = \mu_Q \implies P = Q$$

The Gaussian kernel and many translation-invariant kernels with full support are characteristic.

## 10 Maximum Mean Discrepancy (MMD)

### 10.1 Definition

**Definition 10.1** (Maximum Mean Discrepancy). The MMD between two probability measures  $P$  and  $Q$  with a kernel  $k$  is:

$$\text{MMD}_k(P, Q) := \|\mu_P - \mu_Q\|_{H_k}$$

Equivalently:

$$\text{MMD}_k^2(P, Q) = \mathbb{E}_{x, x' \sim P}[k(x, x')] - 2\mathbb{E}_{x \sim P, y \sim Q}[k(x, y)] + \mathbb{E}_{y, y' \sim Q}[k(y, y')]$$

MMD provides a metric on the space of probability measures via an RKHS.

### 10.2 Witness Function and Interpretation

The function that witnesses the maximum discrepancy—the one that achieves the supremum in the definition—is the normalized difference of mean embeddings:

$$f^*(x) = \frac{\mu_P(x) - \mu_Q(x)}{\|\mu_P - \mu_Q\|_{H_k}}$$

This function shows where the distributions differ most in the feature space.

### 10.3 Two-Sample Testing with MMD

**Proposition 10.1** (Hypothesis Testing). Under the null hypothesis  $H_0 : P = Q$ , the statistic  $\text{MMD}_k^2(P_n, Q_m)$  converges to 0. The asymptotic distribution can be derived for test construction.

For large samples and characteristic kernels, MMD provides a consistent test for distributional differences.

## 11 Hilbert-Schmidt Independence Criterion (HSIC)

### 11.1 Constrained Covariance

**Definition 11.1** (Constrained Covariance). For two probability measures  $P$  on spaces with kernels  $k$  and  $\ell$ , the constrained covariance is:

$$\text{COCO}(P) := \sup_{\|f\|_{H_k}=1, \|g\|_{H_\ell}=1} \text{Cov}[f(X), g(Y)]$$

$$\text{where } \text{Cov}[f(X), g(Y)] = \mathbb{E}[f(X)g(Y)] - \mathbb{E}[f(X)]\mathbb{E}[g(Y)].$$

## 11.2 HSIC Definition

**Definition 11.2** (Hilbert-Schmidt Independence Criterion). The HSIC between random variables  $X$  (with kernel  $k$ ) and  $Y$  (with kernel  $\ell$ ) is:

$$\text{HSIC}(P_{XY}; H_k, H_\ell) := \|\mathcal{C}_{XY}\|_{HS}^2$$

where  $\mathcal{C}_{XY}$  is the cross-covariance operator in the tensor product Hilbert space.

HSIC measures statistical dependence and is zero if and only if  $X \perp Y$ .

## 11.3 Empirical HSIC

For samples  $\{(x_i, y_i)\}_{i=1}^n$ , the empirical HSIC can be computed efficiently as:

$$\widehat{\text{HSIC}} = \frac{1}{n^2} \text{tr}(KHLH)$$

where  $K$  and  $\ell$  are the kernel matrices and  $H = I - \frac{1}{n}\mathbf{1}\mathbf{1}^T$  is the centering matrix.

## Part IV

# Advanced Topics

## 12 Kernel Stein Discrepancy

### 12.1 Stein Operators

**Definition 12.1** (Langevin Stein Operator). For a probability distribution  $p$  with score function  $\nabla \log p(x)$ , the Langevin Stein operator is:

$$(\mathcal{A}_p f)(x) := \nabla \log p(x) \cdot f(x) + \nabla f(x)$$

This operator has the key property that:

$$\mathbb{E}_p[\mathcal{A}_p f] = 0$$

for all appropriate test functions  $f$ .

### 12.2 Definition of KSD

**Definition 12.2** (Kernel Stein Discrepancy). The Kernel Stein Discrepancy (KSD) measures the maximum violation of the Stein equation:

$$\text{KSD}(Q, p; H_k) := \sup_{\|f\|_{H_k} \leq 1} |\mathbb{E}_{X \sim Q}[(\mathcal{A}_p f)(X)]|$$

KSD is zero if and only if  $Q = p$ , making it a proper divergence for distribution testing without density ratio estimation.

KSD is particularly useful when we have an unnormalized density or samples from a distribution but cannot compute the likelihood.

## 13 Optimization and Convex Analysis

### 13.1 Lagrangian and Duality

**Definition 13.1** (Lagrangian). For a constrained optimization problem:

$$\min_x f_0(x) \quad \text{subject to } f_i(x) \leq 0, \quad i = 1, \dots, m, \quad h_j(x) = 0, \quad j = 1, \dots, p$$

the Lagrangian is:

$$L(x, \lambda, \nu) := f_0(x) + \sum_{i=1}^m \lambda_i f_i(x) + \sum_{j=1}^p \nu_j h_j(x)$$

where  $\lambda_i \geq 0$  are inequality constraint multipliers and  $\nu_j$  are equality constraint multipliers.

The dual problem maximizes:

$$g(\lambda, \nu) := \inf_x L(x, \lambda, \nu)$$

subject to  $\lambda \geq 0$ .

## 13.2 Strong Duality and KKT Conditions

**Theorem 13.1** (Strong Duality via Slater's Condition). If the primal problem is convex and Slater's condition holds (there exists a feasible point with strict inequality for inequality constraints), then strong duality holds: the optimal value of the primal equals the optimal value of the dual.

**Theorem 13.2** (KKT Conditions). A point  $x^*$  with multipliers  $(\lambda^*, \nu^*)$  satisfies the KKT conditions if:

1. **Stationarity:**  $0 \in \partial f_0(x^*) + \sum_i \lambda_i^* \partial f_i(x^*) + \sum_j \nu_j^* \partial h_j(x^*)$
2. **Complementary slackness:**  $\lambda_i^* f_i(x^*) = 0$  for all  $i$
3. **Primal feasibility:**  $f_i(x^*) \leq 0$  and  $h_j(x^*) = 0$
4. **Dual feasibility:**  $\lambda_i^* \geq 0$  for all  $i$

For convex problems with strong duality, the KKT conditions characterize optimal solutions.

**Complementary slackness** is key: if  $\lambda_i^* > 0$ , then  $f_i(x^*) = 0$  (the constraint is active).